

Database Architecture — Best Invest Properties

Version: 1.3

Date: 30 September 2026

Storage: Bubble database

Participating systems: Bubble, Make, OpenAI API, SendGrid

1. Purpose and principles

The model must support the three real MVP loops:

- an investor finds, compares, saves and enquires about a specific available unit;
- a developer gets verified, submits a project and maintains price/availability;
- an admin checks documents, publishes content and controls introductions, the score and the AI narrative.

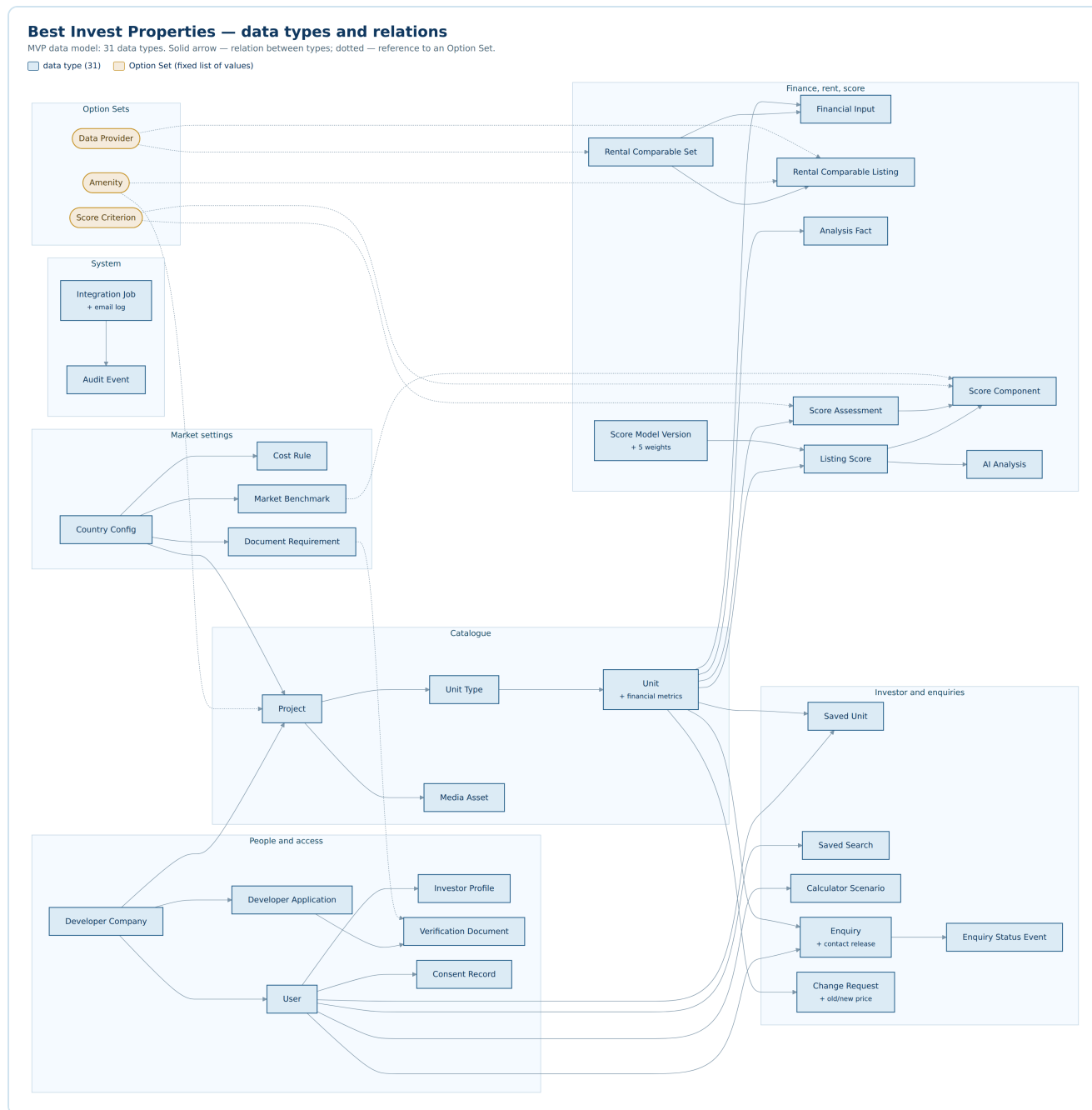
There is one database — Bubble. It holds **31 data types** and **21 Option Sets**.

Principles:

1. **Bubble is the source of truth.** The Make Data Store is not a business database.
2. **Unit is the public listing object.** Project describes the building/complex, Unit Type the repeated configuration, Unit the specific offer.
3. **A single property does not get its own table.** It is a project with one type and one unit.
4. **Fixed calculations are deterministic.** Bubble applies the fixed formulas for acquisition cost, yields, points and the total score, always the same way, to approved values only. Inputs and assessments — for example rent, operating costs, purchase costs and the category assessments — may be proposed with AI assistance from documented sources. They are stored as pending and used only after the admin approves them.
5. **History only where it is needed:** the score model and scores, user consents, and the sources of financial figures. Other changes overwrite the value, and the Audit Event records who changed it and when.
6. **Privacy by default.** New types are created private; only whitelisted fields of published listings are public.
7. **Minimal denormalisation for Bubble.** Fields needed by privacy rules and frequent searches are duplicated on the protected record.
8. **Idempotent integrations.** Every asynchronous process has an Integration Job and a unique idempotency key.

Bubble privacy rules run on the server and must be the primary mechanism that keeps data out of the browser. Visually hiding a group or button is not access control.

2. Conceptual schema



3. Naming rules in Bubble

- Data types: singular English, Pascal Case: Developer Company, Score Model Version.
- Fields: snake_case English: publication_status, price_eur.
- Option sets: Pascal Case: Project Status; options are stable lower-case machine keys.
- Every business type has public_id (text for URLs/references), and is_archived where the record can be archived.
- The Bubble unique id is used internally only; public_id is what goes outside.
- Money in the MVP: numeric fields with the _eur suffix; the currency is always EUR.
- Percentages: store the decimal fraction (0.0725), display as 7.25%.
- Time: all timestamps in UTC; timezone is used only for display.

- Bubble adds `Created Date`, `Modified Date` and `Creator` automatically — they are not repeated in the tables below.

4. Option Sets

Option Sets are for fixed, rarely changing, non-secret values. Bubble states explicitly that option sets are part of the app code, are not protected by privacy rules and are unsuitable for sensitive data. Changing them requires a deploy.

Option Set	MVP values
User Role	<code>investor</code> , <code>developer</code> , <code>admin</code>
Account Status	<code>pending_email</code> , <code>active</code> , <code>suspended</code> , <code>closed</code>
Application Status	<code>draft</code> , <code>submitted</code> , <code>under_review</code> , <code>more_info_required</code> , <code>approved</code> , <code>rejected</code> , <code>withdrawn</code>
Project Status	<code>draft</code> , <code>submitted</code> , <code>under_review</code> , <code>changes_requested</code> , <code>approved</code> , <code>published</code> , <code>paused</code> , <code>sold_out</code> , <code>archived</code> , <code>rejected</code>
Publication Status	<code>not_published</code> , <code>published</code> , <code>hidden</code> , <code>archived</code>
Unit Availability	<code>available</code> , <code>reserved</code> , <code>sold</code> , <code>withdrawn</code>
Financial Input Source	<code>developer_claim</code> , <code>portal_comparable</code> , <code>ai_proposed</code> , <code>analyst_verified</code> , <code>platform_default</code> , <code>calculated</code>
Analysis Fact Tag	<code>source</code> , <code>developer</code> , <code>estimate</code> , <code>gap</code>
Change Request Status	<code>submitted</code> , <code>queried</code> , <code>approved</code> , <code>rejected</code> , <code>applied</code> , <code>cancelled</code>
Enquiry Status	<code>submitted</code> , <code>screening</code> , <code>on_hold</code> , <code>declined</code> , <code>approved_for_intro</code> , <code>introduced</code> , <code>developer_responded</code> , <code>qualified</code> , <code>closed_won</code> , <code>closed_lost</code> , <code>cancelled</code>
Score Verdict	<code>strong</code> , <code>good</code> , <code>watch</code> , <code>high_risk</code> , <code>not_scored</code>
Review Status	<code>not_required</code> , <code>pending</code> , <code>approved</code> , <code>changes_requested</code> , <code>rejected</code>
Analysis Status	<code>queued</code> , <code>generating</code> , <code>generated</code> , <code>failed</code> , <code>stale</code> , <code>published</code>
Job Status	<code>queued</code> , <code>processing</code> , <code>succeeded</code> , <code>retry_wait</code> , <code>failed</code> , <code>dead_letter</code> , <code>cancelled</code>
Document Kind	<code>company_registration</code> , <code>licence</code> , <code>ownership</code> , <code>planning</code> , <code>title</code> , <code>brochure</code> , <code>floor_plan</code> , <code>legal</code> , <code>other</code>
Strategy	<code>long_term_rental</code> , <code>short_term_rental</code> , <code>capital_growth</code> , <code>mixed</code>
Score Criterion	5 criteria — see below
Amenity	<code>communal_pool</code> , <code>gym</code> , <code>gated_area</code> , <code>underground_parking</code> , <code>tennis_golf</code> , <code>concierge</code> , <code>lift</code> , <code>landscaped_gardens</code>
Data Provider	approved rental data sources; attributes: <code>name</code> , <code>country</code> , <code>access_method</code> (<code>api</code> / <code>feed</code> / <code>manual</code>), <code>terms_reference</code> , <code>is_active</code>
Media Kind	<code>photo</code> , <code>floor_plan</code> , <code>brochure</code> , <code>document</code>
Consent Type	<code>terms</code> , <code>privacy</code> , <code>marketing</code> , <code>contact_sharing</code>

What is not an Option Set. Country Config, Cost Rule, Document Requirement and Market Benchmark are data types, because their values change without a deploy. There is no separate settings screen for them in the MVP. Legal texts (Terms, Privacy, Cookie, disclaimers) are static pages; the version the user agreed to is stored on the Consent Record.

Unit Availability . A developer may set only `available` , `reserved` or `sold` , and every such change goes through a Change Request with approval. Only an admin sets `withdrawn` . This must be enforced in the backend workflow whitelist, not in the UI.

Score Criterion — attributes

key	label	max_points	default_weight	Note
<code>income</code>	Rental Income & Net Yield	30	0.30	rating from the net yield table (Project Specification 6.3)
<code>demand</code>	Rental Demand & Tenant Quality	20	0.20	assessed against the level descriptions; may be proposed with AI assistance, approved by the admin
<code>value</code>	Purchase Value & Market Position	20	0.20	rating from the table of total acquisition cost versus the approved estimated fair market value
<code>growth</code>	Growth & Resale Potential	15	0.15	assessed; components: capital appreciation 50%, rental appreciation 25%, resale liquidity 25%
<code>risk</code>	Risk & Investor Protection	15	0.15	assessed; components: legal/owner protection 35%, property/operational costs 25%, regulatory risk 20%, physical/market/security risks 20%; more points = lower risk, scale label mandatory in the UI

Each criterion gets a `rating` from 0 to 10; `weighted_points` = `rating / 10 × weight × 100` . The total score is the sum of the five criterion points (maximum 100). The rating tables and level descriptions are in the Project Specification (6.3). The label of `risk` in the original scoring proposal is "Risk, Costs & Investor Protection"; the interface keeps "Risk & Investor Protection".

- **Provided:** the weights; the tables for `income` and `value` ; the level descriptions for `demand` and `growth` ; the components and their shares for `growth` and `risk` .
- **Still open:** scoring guidelines for which evidence leads to a higher or lower rating for `demand` , `growth` and `risk` ; level descriptions for `risk` ; the score thresholds for the `Score Verdict` labels.
- **Later:** showing sub-scores, tenant-profile percentages and a numerical Data Confidence score in the interface.

Role-specific Enquiry status labels

The eleven `Enquiry Status` values are **canonical**. Role labels are three display attributes of the same Option Set (`label_admin` , `label_investor` , `label_developer`), not additional statuses.

Canonical status	Admin	Investor	Developer
<code>submitted</code>	New	Request received	not shown
<code>screening</code>	Qualifying	Awaiting analysis	not shown
<code>qualified</code>	Hot lead / Awaiting you	Awaiting admin approval	Awaiting admin approval
<code>on_hold</code>	On hold	Under review	On hold
<code>declined</code>	Declined	Not available	Filtered out by Best Invest

Canonical status	Admin	Investor	Developer
approved_for_intro	Connecting	Introduction being prepared	Introduction being prepared
introduced	Connected	Developer contacted	Introduced
developer_responded	In progress	Developer responded	Contacted
closed_won	Converted	Completed	Converted
closed_lost	Closed	Closed	Closed
cancelled	Cancelled	Cancelled	Cancelled

An empty label means the record is not shown to that role at all — this is enforced by a privacy rule, not just hidden in the UI.

AWAITING YOU is not a status. It is the admin label meaning a lead in `qualified` is waiting for their decision. Likewise, "Hot lead" is a label for the same status, not a separate stage.

5. Data types

31 types in six groups.

Group	Types
5.1 People and access	User, Investor Profile, Developer Company, Developer Application, Verification Document, Consent Record
5.2 Market settings	Country Config, Cost Rule, Document Requirement, Market Benchmark
5.3 Catalogue	Project, Unit Type, Unit, Media Asset
5.4 Finance, rent, score	Financial Input, Rental Comparable Set, Rental Comparable Listing, Analysis Fact, Score Model Version, Score Assessment, Listing Score, Score Component, AI Analysis
5.5 Investor and enquiries	Saved Unit, Saved Search, Calculator Scenario, Enquiry, Enquiry Status Event, Change Request
5.6 System	Integration Job, Audit Event

5.1 People and access

User (Bubble built-in)

Field	Type	Required	Note
public_id	text	yes	<code>usr_...</code> , not the email
full_name	text	yes	do not use as an identifier
phone_e164	text	no	private
country_of_residence	text	no	private
roles	list of User Role	yes	at least one role after activation
account_status	Account Status	yes	route guard
email_verified	yes/no	yes	false by default
developer_company	Developer Company	no	for developers; one user per company in the MVP

Field	Type	Required	Note
admin_permission_keys	list of text	no	admin only; permissions are checked by key, not just by role
last_login_at	date	no	audit

Do not store the password in a custom field. Use Bubble authentication.

Investor Profile

Field	Type	Required	Note
user	User	yes	one active profile per User
budget_min_eur	number	no	
budget_max_eur	number	no	
target_gross_yield	number	no	decimal fraction
target_net_yield	number	no	decimal fraction
time_horizon_months	number	no	
countries	list of Country Config	no	preferred countries
strategies	list of Strategy	no	
bedrooms_min	number	no	
marketing_email_opt_in	yes/no	yes	mirrors the latest marketing consent for fast filtering

Profile uniqueness is checked by a backend workflow before creation.

Developer Company

Field	Type	Required	Note
public_id	text	yes	
legal_name	text	yes	
registration_number	text	yes	
registration_country	Country Config	yes	
licence_number	text	yes	
website_url	text	no	
years_active	number	no	
projects_completed	number	no	
projects_selling	number	no	
typical_unit_price_eur	number	no	
verification_status	Application Status	yes	mirrored from the latest application for privacy rules
verified_at	date	no	
is_suspended	yes/no	yes	

The Company owns the Project. Never link a Project directly to a single developer User.

Developer Application

Field	Type	Required	Note
public_id	text	yes	APP-0142
company	Developer Company	yes	
submitted_by_user	User	yes	
status	Application Status	yes	
markets	list of Country Config	yes	
submitted_at	date	no	
decided_at	date	no	
decision_reason_private	text	no	admin only
decision_message_public	text	no	visible to the developer

Verification Document

Field	Type	Required	Note
application	Developer Application	yes	
company	Developer Company	yes	duplicated for privacy rules
document_kind	Document Kind	yes	
file	file	yes	always private
expires_at	date	no	
verification_status	Review Status	yes	
review_note_public	text	no	visible to the developer
reviewed_at	date	no	
is_current	yes/no	yes	a re-upload sets the previous one to no

The file is always private and attached to the Verification Document. Removing the URL does not delete the file; the delete workflow must first run the Bubble action "Delete an uploaded file".

Consent Record

Field	Type	Required	Note
user	User	yes	
consent_type	Consent Type	yes	
granted	yes/no	yes	
document_version	text	yes	version of the text the user agreed to, e.g. terms-2026-09
source_page	text	no	where consent was given
withdrawn_at	date	no	

Consent is never overwritten: every change is a new record. This is a GDPR requirement — you must be able to prove what exactly the user agreed to.

5.2 Market settings

Country Config

Field	Type	Required	Note
iso2	text	yes	CY , ES
name	text	yes	
is_live	yes/no	yes	
coming_soon	yes/no	yes	for Greece/Portugal if needed
default_vacancy_rate	number	yes	decimal fraction
default_management_fee_rate	number	yes	
default_maintenance_rate	number	yes	
default_insurance_annual_eur	number	yes	
legal_disclaimer	text	no	

Cost Rule

Field	Type	Required	Note
country	Country Config	yes	
rule_key	text	yes	transfer_tax , stamp_duty , legal_fees , ...
label	text	yes	
applies_to_new_build	yes/no	no	empty = applies to all
calculation_method	text	yes	flat / percent_price / banded
rate	number	no	for percent_price
fixed_amount_eur	number	no	for flat
bands_json	text	no	for banded
source_url	text	yes	official source
source_checked_at	date	yes	
approved_by	text	no	who checked the rate (consultant)

Do not store all tax rules as one text. When a value changes, it is overwritten and the Audit Event records the change history.

Document Requirement

Field	Type	Required	Note
country	Country Config	yes	
subject_type	text	yes	developer / project
document_kind	Document Kind	yes	
required_for_account_creation	yes/no	yes	e.g. company registration, licence
required_for_submission	yes/no	yes	
required_for_publication	yes/no	yes	e.g. building permit, escrow
expiry_months	number	no	empty = no expiry

Field	Type	Required	Note
instructions	text	no	hint for the developer
is_active	yes/no	yes	

The three `required_for_*` fields express the three levels on the application screen: `REQUIRED` / `TO CONFIRM` / `OPTIONAL`. The matrix is **fully configurable** by the admin per country and is not hard-coded.

Market Benchmark

Reference data on comparable properties, e.g. the median asking price per m² in a district. It can support the estimated fair market value used for `Purchase Value & Market Position`, together with the comparables for the specific property.

Field	Type	Required	Note
country	Country Config	yes	
city	text	yes	
area_name	text	no	empty = whole city
property_type	text	no	
median_price_per_m2_eur	number	yes	median for new builds
sample_size	number	no	number of properties in the sample
source_reference	text	yes	public listings / transaction register
checked_at	date	yes	date checked

5.3 Property catalogue

Project

Field	Type	Required	Note
public_id	text	yes	PRJ-0311
company	Developer Company	yes	never returned to an anonymous request
country	Country Config	yes	
city	text	yes	
area_name	text	no	
address_private	text	no	not published
map_lat_rounded	number	no	rounded for the public map
map_lng_rounded	number	no	
name	text	yes	
project_type	text	yes	single / complex
description_public	text	no	
completion_date	date	yes	
declared_unit_count	number	yes	
amenities	list of Amenity	no	
cover_media	Media Asset	no	required for publication

Field	Type	Required	Note
status	Project Status	yes	
publication_status	Publication Status	yes	
published_at	date	no	

Country/city/area are enough for search cards; `address_private` is never exposed publicly.

Developer anonymity. The developer's name and contacts are not shown in the catalogue or on the property page; the public label is "Introduced by Best Invest". The privacy rule on Project does not return `company` to an anonymous request at all — do not rely on the UI not showing it.

Unit Type

Field	Type	Required	Note
public_id	text	yes	
project	Project	yes	
name	text	yes	e.g. "Type B — 2 bed"
bedrooms	number	yes	
bathrooms	number	yes	
indoor_area_m2	number	yes	
outdoor_area_m2	number	no	
floor_plan	Media Asset	no	

Unit

The central listing record. It also holds the current financial metrics — there is no separate type for them.

Field	Type	Required	Note
public_id	text	yes	URL and external references
project	Project	yes	
unit_type	Unit Type	yes	
unit_number	text	yes	may be hidden until introduction
floor	number	no	
price_eur	number	yes	current approved price
availability	Unit Availability	yes	
has_pending_change	yes/no	yes	a Change Request is in progress — the portal row shows "pending approval"
monthly_rent_eur	number	no	approved rent used for scoring
purchase_costs_eur	number	no	calculated from approved inputs
acquisition_cost_eur	number	no	calculated: price + purchase costs
annual_operating_costs_eur	number	no	calculated from approved inputs
annual_net_income_eur	number	no	calculated

Field	Type	Required	Note
gross_yield	number	no	calculated
net_yield	number	no	calculated
financials_calculated_at	date	no	date of the last calculation
investment_score	number	no	current score
score_verdict	Score Verdict	no	badge
current_score	Listing Score	no	components and model version
country_cached	Country Config	yes	for search and privacy without a deep chain
city_cached	text	yes	search
bedrooms_cached	number	yes	search
area_m2_cached	number	yes	search
property_type_cached	text	yes	search
completion_date_cached	date	yes	completion filter
strategy_keys	list of Strategy	no	search
publication_status	Publication Status	yes	only <code>published</code> is public

Financial and cached fields are updated by the atomic backend workflow `recalculate_unit`. Do not build the public search over many nested relations.

Base MVP formulas:

```

annual_gross_rent = monthly_rent × 12
annual_effective_rent = annual_gross_rent × (1 - vacancy_rate)
gross_yield = annual_gross_rent / price
annual_net_income = annual_effective_rent
    - management - maintenance - insurance - property_tax - other_costs
acquisition_cost = price + purchase_costs
net_yield = annual_net_income / acquisition_cost

```

Inputs are approved Financial Inputs (rent comes from reviewed comparable rentals), plus the Country Config settings and Cost Rules for the country. The values used for a particular score are recorded on the Listing Score.

Media Asset

Field	Type	Required	Note
project	Project	yes	
unit_type	Unit Type	no	for floor plans
kind	Media Kind	yes	
image	image	no	for <code>photo</code>
file	file	no	for PDFs/documents
is_private	yes/no	yes	legal documents — <code>yes</code>
is_cover	yes/no	yes	
sort_order	number	yes	

Field	Type	Required	Note
caption	text	no	

Public photos/brochures and private documents are different records with different privacy rules.

5.4 Finance, rent, score

Financial Input

The Project Review screen shows a panel where every figure has its own source and assumptions, and three rent values side by side (claimed by the developer, Best Invest's comparable estimate, and the one actually used for scoring). So each input value is a separate record.

Field	Type	Required	Note
unit	Unit	yes	
input_key	text	yes	monthly_rent, annual_operating_costs, purchase_costs, vacancy_allowance, estimated_fair_market_value, claimed_yield, ...
label	text	yes	label in the review
value_number	number	no	empty when the item is marked as a gap
is_gap	yes/no	yes	no source supports a value
source	Financial Input Source	yes	
source_reference	text	no	"Larnaca district, 5 comparable lettings"
sources_json	text	no	the sources used: id, link or document, date
assumptions	text	no	required for ai_proposed
reliability_note	text	no	required for estimated_fair_market_value: how reliable the evidence is
comparable_set	Rental Comparable Set	no	required for portal_comparable
integration_job	Integration Job	no	the AI request that produced an ai_proposed value
is_used_for_scoring	yes/no	yes	only one active value per input_key
review_status	Review Status	yes	pending until approved by the admin
approved_at	date	no	
superseded_by	Financial Input	no	a new value does not overwrite the old one

Rules:

- only values with review_status = approved reach the public listing;
- the Unit's financial metrics are calculated **only** from approved Financial Inputs;
- a value with source = developer_claim can never have is_used_for_scoring = yes without separate admin approval;
- a value with source = ai_proposed must have at least one source in sources_json and its assumptions; it is pending until the admin approves it. A correction by the admin is saved as a new record with source = analyst_verified;
- a gap (is_gap = yes) never has a value and can never be used for scoring;

- `source = portal_comparable` requires a `comparable_set` with `review_status = approved`;
- for rent there are always at least two records: the developer's claim (`developer_claim`) and the value derived from comparable listings (`portal_comparable`) — these are shown side by side on Project Review;
- `developer_claim` never turns into `portal_comparable` automatically;
- a record is not edited: a new value creates a new record and sets `superseded_by` on the old one. This shows where every published figure came from.

Rental Comparable Set

One sample of comparable rentals for one parameter set on a given date. The individual listings are stored as Rental Comparable Listing records, so every figure can be traced back to its sources.

Field	Type	Required	Note
public_id	text	yes	
retrieved_at	date	yes	date the sample was put together
country	Country Config	yes	matching parameter
city	text	yes	matching parameter
area_name	text	no	district
property_type	text	yes	matching parameter
bedrooms	number	yes	matching parameter
area_m2_min	number	no	sample floor-area bound
area_m2_max	number	no	sample floor-area bound
features	list of Amenity	no	relevant features used to assess the match, e.g. pool, parking, gated area
listings_count	number	yes	calculated by Bubble from the linked listings
rent_min_eur	number	yes	calculated by Bubble
rent_median_eur	number	yes	calculated by Bubble
rent_max_eur	number	yes	calculated by Bubble
collection_method	text	yes	<code>api</code> / <code>feed</code> / <code>manual_import</code> / <code>manual_entry</code>
integration_job	Integration Job	no	empty for manual entry
review_status	Review Status	yes	pending until checked by the admin
is_current	yes/no	yes	current sample for this parameter set

Rules:

- the count, range and median are technical processing of the sample by Bubble, not an approved figure;
- a sample with `review_status ≠ approved` cannot be a source for a Financial Input and is not shown to the investor;
- relevant features are factors in assessing the match; a comparable does not have to match on every one;
- a new sample does not overwrite the previous one: the old one gets `is_current = no`;

- where there is no suitable source, the admin enters documented comparables by hand (`manual_entry`) — they are stored the same way. Otherwise the estimate stays `pending`; no rent estimate is ever invented.

The access method and the right to use each source's data are checked before it is connected. The link to the terms is the `terms_reference` attribute in the `Data Provider Option Set`.

Rental Comparable Listing

One comparable rental in a sample, with its source.

Field	Type	Required	Note
<code>public_id</code>	text	yes	source id used in the analysis, e.g. <code>src_01</code>
<code>comparable_set</code>	Rental Comparable Set	yes	
<code>provider</code>	Data Provider	no	Option Set; empty for a manually documented comparable
<code>source_url</code>	text	no	link to the listing; required unless a document is attached
<code>source_document</code>	file	no	private; evidence for a manually entered comparable
<code>retrieved_at</code>	date	yes	date the listing was retrieved
<code>monthly_rent_eur</code>	number	yes	
<code>property_type</code>	text	yes	
<code>bedrooms</code>	number	yes	
<code>area_name</code>	text	no	
<code>indoor_area_m2</code>	number	no	
<code>features</code>	list of Amenity	no	where this data is available

Every listing has a link or a document and a date, so each comparable behind the range and the median can be checked. The records are not edited; a new collection creates a new sample.

Analysis Fact

The "Sources, assumptions & gaps" block on the analysis and review screens.

Field	Type	Required	Note
<code>unit</code>	Unit	yes	
<code>tag</code>	Analysis Fact Tag	yes	
<code>body</code>	text	yes	fact text
<code>source_url</code>	text	no	
<code>sort_order</code>	number	yes	
<code>is_active</code>	yes/no	yes	

The `gap` tag marks a known absence of data and must be visible to the investor — it is a legally significant part of the disclosure.

Score Model Version

Field	Type	Required	Note
version_number	number	yes	v1, v2, ...
status	text	yes	draft / active / retired
weight_income	number	yes	decimal fraction, default 0.30
weight_demand	number	yes	default 0.20
weight_value	number	yes	default 0.20
weight_growth	number	yes	default 0.15
weight_risk	number	yes	default 0.15
change_reason	text	yes	shown in the versions table
properties_rescored	number	no	filled in after recalculation
activated_at	date	no	

Weight rules:

- the five weights of the active model add up to **1.0** ; until they do, Save is blocked (checked by a backend workflow);
- the editor works in percentages, 0-100 per criterion; there is **no** hard cap on a single criterion;
- the UI shows a soft warning if any criterion exceeds **50%** — one criterion would then effectively determine the whole score. The warning does not block saving; the threshold is a UI constant;
- the weights are the same for Cyprus and Spain;
- **Revert** creates a **new** version with the previous parameters rather than deleting history.

Score Assessment

A proposed or approved assessment for one of the three assessed criteria (**demand** , **growth** , **risk**) of a unit. **income** and **value** are not assessed: Bubble derives their ratings from the tables. It exists **before** any score is calculated, so a proposal or a gap can be stored and reviewed without a Listing Score.

Field	Type	Required	Note
public_id	text	yes	
unit	Unit	yes	
criterion	Score Criterion	yes	only demand , growth , risk
rating	number	no	0-10; empty for a gap
component_ratings_json	text	no	for growth and risk : the rating of each component (0-10)
is_preliminary	yes/no	yes	the evidence is insufficient; the investor sees a disclaimer
is_gap	yes/no	yes	no source supports an assessment
explanation	text	no	required unless it is a gap
assumptions	text	no	

Field	Type	Required	Note
sources_json	text	no	the sources used: id, link or document, date; required unless it is a gap
source	Financial Input Source	yes	ai_proposed or analyst_verified
integration_job	Integration Job	no	the AI request that produced an ai_proposed assessment
review_status	Review Status	yes	pending until the admin approves it
approved_at	date	no	
superseded_by	Score Assessment	no	a correction or a new proposal does not overwrite the old one

Rules:

- an AI proposal is always stored as pending; the admin approves it or saves a correction as a new record with source = analyst_verified;
- a gap can never be approved as a rating;
- per unit and criterion only one approved, not superseded assessment is used;
- for growth and risk, the rating combines the component ratings in the component shares (for example 8, 7 and 9 for growth give 8.0);
- Bubble calculates a Listing Score only when demand, growth and risk have an approved assessment and the estimated fair market value is approved.

Listing Score

Field	Type	Required	Note
unit	Unit	yes	
model_version	Score Model Version	yes	
total_score	number	yes	0-100
verdict	Score Verdict	yes	the score thresholds for the labels are still to be agreed
is_preliminary	yes/no	yes	yes if any assessment used is preliminary
price_eur_used	number	yes	price at calculation time
monthly_rent_eur_used	number	yes	rent at calculation time
net_yield_used	number	yes	
gross_yield_used	number	yes	
calculated_at	date	yes	
is_current	yes/no	yes	a new calculation sets the previous one to no

The *_used fields record the inputs the score was calculated from, so the score stays reproducible even after the price or rent on the Unit changes.

Score Component

Created by Bubble when the score is calculated, from approved values only.

Field	Type	Required	Note
listing_score	Listing Score	yes	
criterion	Score Criterion	yes	Option Set
assessment	Score Assessment	no	the approved assessment used; required for demand , growth , risk
raw_value	number	no	net yield for income ; position versus estimated fair market value for value ; empty for assessed criteria
rating	number	yes	0-10: from the tables for income and value , copied from the approved assessment for the others
weighted_points	number	yes	$\text{rating} / 10 \times \text{weight} \times 100$
explanation	text	no	copy of the approved explanation at calculation time
sources_json	text	no	copy of the approved sources at calculation time

Each Listing Score has exactly five components; they add up to the total score within 0.01 . The copies keep the score reproducible even if a later assessment supersedes the one used. The approved explanation and sources are passed to OpenAI for the analysis text, so the text can explain the points without guessing.

AI Analysis

Field	Type	Required	Note
unit	Unit	yes	
listing_score	Listing Score	yes	the score the text is based on
status	Analysis Status	yes	
review_status	Review Status	yes	
prompt_version	text	yes	
model_id	text	yes	
headline	text	no	analysis headline
summary	text	no	
score_explanation	text	no	explanation of the score
strengths_json	text	no	
risks_json	text	no	
missing_data_json	text	no	
input_tokens	number	no	
output_tokens	number	no	
reviewed_at	date	no	
published_at	date	no	
failure_code	text	no	

Only `review_status = approved` is published, and only while the linked Listing Score is still `is_current = yes`.

AI Analysis holds only the analysis text (the second AI stage). Proposed inputs and assessments from the first stage are stored as Financial Inputs (`source = ai_proposed`) and Score Assessments (`review_status = pending`).

5.5 Investor and enquiries

Saved Unit

Field	Type	Required	Note
user	User	yes	
unit	Unit	yes	
price_at_save_eur	number	yes	for the "price changed" badge
availability_at_save	Unit Availability	yes	for the "status changed" badge
is_active	yes/no	yes	check for an active duplicate before creating

Saved Search

Field	Type	Required	Note
public_id	text	yes	
user	User	yes	
name	text	yes	
countries	list of Country Config	no	
property_types	list of text	no	
bedrooms	list of text	no	<code>studio</code> , <code>1</code> , <code>2</code> , <code>3+</code>
strategies	list of Strategy	no	
completion	list of text	no	<code>ready</code> , <code>lt_12m</code> , <code>12_24m</code>
price_max_eur	number	no	empty = no upper limit (the slider's top position, €600k+)
gross_yield_min	number	no	
net_yield_min	number	no	
is_active	yes/no	yes	

When a new property matching an active saved search is published, Bubble sends the investor an email through an Integration Job (`template_key = saved_search_match`); the idempotency key prevents a second email for the same property and search.

Calulator Scenario

Field	Type	Required	Note
public_id	text	yes	
user	User	yes	anonymous calculations are not stored
unit	Unit	yes	
name	text	yes	

Field	Type	Required	Note
strategy	Strategy	yes	long / short term
rent_scenario	text	yes	base / average / best
purchase_price_eur	number	yes	
monthly_rent_eur	number	yes	
occupancy_rate	number	yes	
management_fee_rate	number	yes	
annual_net_income_eur	number	yes	result at save time

There are no mortgage, deposit, interest-rate or loan-term fields in the MVP.

Enquiry

The contact release fields are stored right here — there is no separate type for them.

Field	Type	Required	Note
public_id	text	yes	ENQ-YYYY-... / for the developer LEAD-0412
investor_user	User	yes	private
unit	Unit	yes	the specific subject of the enquiry
project	Project	yes	denormalised
developer_company	Developer Company	yes	denormalised for privacy rules
status	Enquiry Status	yes	canonical status
budget_eur	number	no	value at submission time
consent_to_share_contact	yes/no	yes	separate consent
fit_summary_private	text	no	admin only
follow_up_date	date	no	required for on_hold
submitted_at	date	yes	SLA start
contact_released_at	date	no	filled in on Approve & connect
released_investor_name	text	no	copy the developer sees after release
released_investor_email	text	no	
released_investor_phone	text	no	
released_developer_contact	text	no	developer contact the investor sees
developer_outcome	text	no	lead outcome: viewing_booked / reserved / not_interested
closed_at	date	no	

The developer **never** gets access to the investor's User record. Before release the `released_*` fields are empty; the privacy rule returns them to the developer only once `contact_released_at` is filled in.

Enquiry Status Event

Field	Type	Required	Note
enquiry	Enquiry	yes	
from_status	Enquiry Status	no	
to_status	Enquiry Status	yes	
actor_user	User	no	empty for automatic transitions
reason_private	text	no	required for <code>on_hold</code> / <code>declined</code> ; never shown to the investor
note_investor	text	no	what the investor sees in the status history

A status never changes without a Status Event. These records are what the investor sees as the status history.

Change Request

One request = one change to one unit, or a content review request.

Field	Type	Required	Note
public_id	text	yes	
project	Project	yes	
unit	Unit	no	empty for content review
company	Developer Company	yes	for privacy rules
request_type	text	yes	<code>price</code> / <code>availability</code> / <code>content_review</code>
old_price_eur	number	no	
new_price_eur	number	no	
old_availability	Unit Availability	no	
new_availability	Unit Availability	no	only <code>available</code> / <code>reserved</code> / <code>sold</code>
content_message	text	no	for <code>content_review</code> : what the developer wants to change
status	Change Request Status	yes	
decision_reason	text	no	required for <code>queried</code> / <code>rejected</code> ; visible to the developer
submitted_at	date	yes	
decided_at	date	no	

After publication the developer changes only price and availability; both go through **prior** approval. Locked fields (project name, location, completion date, unit mix, specification, media) change through `content_review`. The Change Request is kept after it is applied — this is the change history on the Project & Units screen.

5.6 System

Integration Job

Every asynchronous action: Make calls and every email sent by Bubble.

Field	Type	Required	Note
public_id	text	yes	correlation id
job_type	text	yes	ai_proposals_generate, ai_analysis_generate, email, introduction, rental_comparables, score_rescore_batch, ...
entity_type	text	yes	
entity_public_id	text	yes	
recipient_user	User	no	for emails
template_key	text	no	for emails
status	Job Status	yes	
idempotency_key	text	yes	unique
attempt_count	number	yes	
processed_count	number	no	batch recalculation progress
last_error_redacted	text	no	no PII or secrets
provider_message_id	text	no	email id / OpenAI response id
completed_at	date	no	

Uniqueness is enforced by `idempotency_key`. The Job is checked before the action (a Make call or sending an email); if it is `succeeded`, the repeat ends without side effects. This same type is the log of sent emails. Failed jobs are logged and the team is notified; there is no separate monitoring screen in the MVP.

Audit Event

Field	Type	Required	Note
actor_user	User	no	empty for system actions
action_key	text	yes	project.published, cost_rule.updated, ...
entity_type	text	yes	
entity_public_id	text	yes	
before_json	text	no	no PII
after_json	text	no	no PII
reason	text	no	
source	text	yes	bubble_ui / bubble_backend / make / openai

Audit Event is append-only; user workflows may not change or delete records. It records who changed settings (rates, weights, documents) and when, instead of keeping separate versions of each record.

6. Status transitions

Project lifecycle

```
draft → submitted → under_review
under_review → changes_requested → submitted
under_review → approved → published
under_review → rejected
published → paused | sold_out | archived
```

The admin performs `approved → published` as a separate action. Publishing is not allowed without approved verification, at least one available unit, a cover photo, calculated financials, a current score, and an approved AI analysis or an explicit deterministic-only fallback.

A developer can **submit** a project with an incomplete set of units (modal: "12 units declared · 3 entered. You can submit and add the rest before publication"). So unit completeness is a condition of `approved → published`, not of `draft → submitted`. In addition, publishing requires every Financial Input used in the score and every Score Assessment used in the score to have `review_status = approved`.

Enquiry decision

UI action	Canonical transition	Emails
Approve & connect	<code>qualified → approved_for_intro → introduced</code>	to both parties
Hold	<code>qualified → on_hold</code>	none
Decline	<code>qualified → declined</code>	investor only, in neutral wording

`on_hold` and `declined` require `reason_private` — for Decline it is a short free-text internal note that neither the investor nor the developer sees. The text the investor sees on a decline comes from a template and does **not** contain the reason. `on_hold` additionally requires `follow_up_date` on the Enquiry.

Developer Application lifecycle

```
draft → submitted → under_review
under_review → more_info_required → submitted
under_review → approved | rejected
submitted/under_review → withdrawn
```

Enquiry lifecycle

```
submitted → screening → qualified → approved_for_intro → introduced
screening | qualified → on_hold | declined
on_hold → qualified | declined
introduced → developer_responded → closed_won | closed_lost
submitted | screening | qualified → cancelled
```

`qualified` means "the lead has been checked by the analyst and is waiting for the admin's decision", so it comes **before** `approved_for_intro`. The stage after the developer responds is `developer_responded`, then `closed_won` / `closed_lost`.

7. Privacy rules matrix

Data type	Anonymous	Investor owner	Developer company member	Admin
User	none	own whitelisted fields	own only	full
Investor Profile	none	own	none	full
Developer Company	none	none	own	full
Developer Application / Verification Document	none	none	own company	full
Consent Record	none	own	own	full
Project	published public subset, without company	published public subset	own company	full
Unit	published + available public subset	same	own company units	full
Media Asset public	published records	published records	own company	full
Media Asset private	none	none	own company	full
Financial Input	none	none	own project, approved only	full
Rental Comparable Set / Rental Comparable Listing	none	none	none	full
Analysis Fact / Listing Score / Score Component	published subset	published subset	own project	full
Score Assessment	none	none	none	full
AI Analysis	approved/published only	approved/published only	own project published view	full
Enquiry	none	own, investor-safe fields	own company; released_* fields only after release	full
Enquiry Status Event	none	own, without reason_private	own company, without reason_private	full
Change Request	none	none	own company	full
Integration Job / Audit Event	none	none	none	full